

When Does Search Beat Reasoning?

A Topology-Driven Diagnostic of A* versus Chain-of-Thought

Anonymous authors

Paper under double-blind review

Abstract

Chain-of-Thought (CoT) prompting and tree-search methods have independently advanced LLM reasoning, yet no systematic study explains *when* search-based reasoning outperforms linear reasoning or vice versa. We treat A* search as a **unified search framework**—subsuming breadth-first, depth-first, and best-first strategies under a single admissible heuristic—and conduct a controlled comparison against CoT on 2,151 questions across three benchmarks (LogicQA, StrategyQA, HotpotQA) using LLaMA 3.1-8B. Aggregate results are equivocal: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). To resolve this ambiguity, we introduce two complementary diagnostic lenses. First, a **Reasoning Topology Framework** characterises each question by 13 structural features and clusters problems into four topology types, showing that branching complexity—not dataset identity—predicts strategy success. Second, **cross-dataset topic modelling** (NMF, 8 categories) reveals that A* dominates in 6 categories covering **66.2%** of the corpus—all requiring multi-hop evidence chaining—while CoT wins on the remaining 33.8% of single-step inference tasks. A topology-aware router that sees *zero model outputs* recovers 26.1% of the oracle gap, outperforming all output-based routers, confirming that the bottleneck is recognising problem structure before inference begins. We further document the Fluency–Correctness Asymmetry (38% of divergent cases), a failure-mode taxonomy (91.1% overthinking), and a $3.5\times$ search-as-capability-amplification effect for small language models. Together, these findings establish that neither strategy universally dominates: the right choice is determined by the reasoning topology of the problem.

Keywords: A* search, chain-of-thought, LLM reasoning, topic modelling, reasoning topology, adaptive routing

1 Introduction

The emergence of Chain-of-Thought (CoT) prompting (Wei et al., 2022) demonstrated that large language models (LLMs) reason more reliably when they externalise intermediate steps. Subsequent work extended this insight from linear chains to richer structures: Tree-of-Thoughts (ToT; Yao et al. 2024) explores alternative reasoning paths via breadth-first or depth-first search, Graph-of-Thoughts (GoT; Besta et al. 2024) permits arbitrary reasoning topologies, and MCTS-based approaches (Hao et al., 2023) apply Monte Carlo rollouts to reasoning trees. These advances raise a fundamental question that remains unanswered:

When does structured search actually improve over linear reasoning, and when does it hurt?

Search-based reasoning relaxes the linearity assumption, but exploration is not free. Search can preserve useful alternatives, but it can also introduce over-exploration, redundant decomposition, spurious branches, and unnecessary reasoning on problems that are already solvable by a compact linear chain. Therefore, the important question is not simply whether search is stronger than CoT, but *when the input contains useful branching structure*.

Prior comparisons treat search-based and linear reasoning as competing paradigms, reporting aggregate accuracy on individual benchmarks. But aggregates conceal systematic structure. A method that wins by 10 points on common-

sense reasoning and loses by 1 point on formal logic looks equivalent on average—yet behaves very differently at the instance level.

We decompose the overarching question into four testable sub-questions:

- (i) Does guided search dominate linear CoT under a controlled comparison?
- (ii) If not, do the two methods succeed on complementary instances?
- (iii) Can this complementarity be explained by pre-generation structural properties of the input?
- (iv) Can those structural properties be used to route between CoT and A* *before* any reasoning trace is generated?

Why A* as a diagnostic framework. We use A* not because it is the universally best search algorithm for LLM reasoning, but because it exposes the minimal ingredients of guided search: a frontier of partial reasoning states, an accumulated path cost, an estimated remaining cost, a branching budget, and a stopping rule. By varying the heuristic, A* reduces to BFS ($h=0$), DFS, greedy best-first ($g=0$), and beam search as special cases. This lets us study the value of *guided exploration itself*—what branching and prioritisation contribute—rather than the idiosyncrasies of any particular search variant. Tree-of-Thoughts and beam-style methods entangle evaluator-based pruning with decomposition prompts and breadth choices; MCTS-style methods add rollout and value-estimation design choices. A* gives a compact diagnostic abstraction for asking when guided exploration is useful.

The inconsistency problem. We evaluate A* and CoT on 2,151 questions across three benchmarks using LLaMA 3.1-8B, and find that aggregate results are **inconsistent**: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). Self-consistency (Wang et al., 2023) and Tree-of-Thoughts fall between the two. No single strategy dominates, and the aggregate numbers provide no actionable guidance for deployment.

Our approach. Rather than declaring a winner, we ask *why* the methods diverge. We bring two diagnostic tools to bear:

1. **Reasoning Topology Framework**—13 structural features (question length, context complexity, branching coefficient, logical connectives, hop count) characterise each problem’s reasoning structure. K-means clustering yields 4 topology types with distinct strategy preferences (§4.3).
2. **Cross-dataset Topic Modelling**—Non-negative Matrix Factorisation (NMF) on 3,660 paired question texts identifies 8 content categories that cut across datasets. Measuring A* versus CoT accuracy within each category reveals that A* dominates in 6 categories covering 66.2% of the corpus (§4.4).

These two lenses converge on a single structural explanation: **A* wins whenever the correct answer requires traversing a chain of two or more interdependent facts, where greedy commitment at hop 1 propagates errors to the final answer.** When only one inference step separates question from answer, CoT’s linear approach is sufficient and A*’s branching adds overhead without benefit.

Contributions.

1. A controlled comparison using A* as a diagnostic framework, with approximate admissibility analysis (three sufficient conditions) and empirical validation (§3.2).
2. A Reasoning Topology Framework with 4 empirically validated cluster types (§4.3).
3. The finding that A* dominates in 66.2% of question categories via cross-dataset topic modelling (§4.4).
4. A topology-aware pre-generation router that outperforms all output-based routers (§4.5).
5. The Fluency–Correctness Asymmetry quantified at 38.0% (§4.6).
6. A failure-mode taxonomy showing 91.1% of A* failures are overthinking (§4.7).
7. Evidence that A* amplifies small language models by $3.5\times$ on multi-hop tasks (§4.8).

Table 1: Methodological comparison of reasoning strategies.

Method	Structure	Frontier?	Heuristic?	Pre-gen. routing?	Main limitation
CoT	linear	no	no	no	premature commitment
Self-Consistency	multi-linear	no	no	no	unguided sampling
ToT	tree/BFS	yes	evaluator	no	breadth/evaluator entangled
MCTS/LATS	tree + rollout	yes	value/rollout	no	many design choices
A* (ours)	priority frontier	yes	yes	yes	higher compute, topology-sensitive

2 Related Work

Chain-of-thought and structured reasoning. CoT (Wei et al., 2022) and self-consistency (Wang et al., 2023) established the value of explicit intermediate reasoning. ToT (Yao et al., 2024) and GoT (Besta et al., 2024) extend to richer structures. CoT-decoding (Wang and Zhou, 2024) shows that reasoning paths exist in top- k alternative tokens without prompting. A recent empirical study finds CoT is not universally beneficial (Meincke et al., 2025); our result is the search-side complement—identifying precisely the structural conditions under which search helps. Our work differs in three respects: A*’s priority rule (vs. undirected BFS/DFS), a formal admissibility theorem (ToT lacks one), and systematic characterisation of *when search hurts*.

Search-augmented reasoning. MCTS-based reasoning (Hao et al., 2023) and self-evaluation beam search (Xie et al., 2024) use search frameworks without complementarity analysis. LATS (Zhou et al., 2024) unifies reasoning, acting, and planning via MCTS. MCTS_r (Zhang et al., 2024) applies MCTS with self-refinement to mathematical Olympiad problems. Coconut (Hao et al., 2024) enables implicit BFS through continuous latent states—our work is the discrete, interpretable counterpart. rStar-Math (Guan et al., 2025) demonstrates search helps SLMs on mathematics; we confirm this in QA and identify the capability–topology interaction that determines the magnitude of the gain (§4.8).

Over-exploration and search failures. The Fetch framework (Zhang et al., 2025a) identifies redundant-state over-exploration; this corresponds to our step repetition failure mode. They frame it as an efficiency problem; we diagnose it as a *correctness* problem. The overthinking survey (Zhang et al., 2025b) documents the phenomenon in long-CoT models; CoT-Valve (Ma et al., 2025) addresses it via length-compressible tuning. Our 91.1% overthinking rate is the first per-instance quantification in search-based reasoning.

Routing and adaptive computation. Routing between strategies is distinct from architectural routing (MoE; Shazeer et al. 2017). A recent study (Meincke et al., 2025) shows CoT is not universally beneficial; our result is the search-side mirror. Prior routing uses output features; our topology router is the first to use pre-generation structural features, recovering more oracle gap.

LLM-based evaluation. LLM-as-a-judge (Zheng et al., 2023) and process supervision (Lightman et al., 2023) motivate our critic-based routing. The Fluency–Correctness Asymmetry explains why critic-based routing reaches a ceiling.

Methodological positioning. Table 1 summarises the structural differences between reasoning strategies relevant to our analysis.

3 Methodology

3.1 A* Search for LLM Reasoning

We formalise reasoning as a graph search problem. Each node n represents a partial reasoning state (the set of intermediate conclusions reached so far). Edges correspond to single reasoning steps generated by the LLM. A* maintains an open list of nodes sorted by $f(n) = g(n) + h(n)$, where $g(n)$ is the accumulated cost (tokens consumed) and $h(n)$ is the heuristic estimate of remaining cost to a correct answer.

State representation. A state s is a tuple $(\mathbf{q}, \mathbf{c}, \mathcal{P})$ where $\mathbf{q}$ is the question, $\mathbf{c}$ is the context, and $\mathcal{P} = \{p_1, \dots, p_k\}$ is the set of premises derived so far.

Expansion. At each step, A* pops the most promising node and generates $N_e = 3$ candidate successors by prompting the LLM to extend the reasoning by one step, each from a different inference angle (*e.g.*, supporting evidence, counterargument, factual lookup).

Heuristic evaluation. A critic model (DeepSeek-R1-Distill-Qwen-14B) scores each successor for (a) factual consistency with the context and (b) estimated reasoning completeness. The critic returns a score in $[0, 1]$ converted to $h(n) = 1 - \text{score}$, so higher-quality states have lower estimated remaining cost. We use DeepSeek-R1-Distill-Qwen-14B because it is a strong open instruction-following model for reasoning-trace critique, is architecturally distinct from the LLaMA-3.1-8B generation backbone (avoiding self-evaluation bias), and can be run locally without closed-API dependence. The critic is treated as an output-based routing baseline in the analysis; the main routing result is the topology router, which uses no LLM inference.

Termination. Search terminates when a node’s reasoning chain reaches a definitive answer, or when a budget of $B = 15$ expansions is exhausted. The highest-scoring terminal node’s answer is returned.

3.2 Admissibility of the LLM Heuristic

A heuristic $h(n)$ is admissible if $h(n) \leq h^*(n)$ for all nodes n , where $h^*(n)$ is the true remaining cost. In classical planning, admissibility ensures A* finds the optimal solution. We identify three sufficient conditions under which the LLM-generated heuristic approximates admissibility:

Proposition 1 (Sufficient Conditions for Heuristic Admissibility). *Let $h(n) = 1 - \text{score}(n)$ be the critic-generated heuristic. Then $h(n)$ approximates admissibility under any of:*

Case 1 (Monotone scoring). *If the critic assigns monotonically increasing scores to states with strictly more correct premises, then $h(n)$ never overestimates. This gives exact admissibility.*

Case 2 (Conservative estimation). *If $\mathbb{E}[\text{score}(n)] \leq \text{score}^*(n)$ (systematic underestimation), then $\mathbb{E}[h(n)] \geq h^*(n)$, making the heuristic admissible in expectation. This weaker guarantee implies A* finds the optimal solution with high probability over repeated runs.*

Case 3 (Bounded overestimation). *If $|\text{score}(n) - \text{score}^*(n)| \leq \epsilon$, then A* finds a solution within ϵ of optimal (ϵ -admissibility; Hart et al. 1968).*

Empirical validation. We annotate 500 reasoning states (100 per dataset for StrategyQA and HotpotQA, 300 for LogicQA). The DeepSeek critic underestimates ground-truth reasoning progress (mean score = 0.43 vs. human-assessed = 0.58; paired t -test $p < 0.001$), satisfying Case 2. The overestimation rate is 12.3% of nodes with maximum $\epsilon_{\max} = 0.14$, providing an empirical bound for Case 3. We note that these conditions are *sufficient but not necessary*—A* can still be effective even when the heuristic is occasionally inadmissible, as the search budget acts as a regulariser.

3.3 Comparison Methods

We compare five reasoning strategies, all using LLaMA 3.1-8B via Ollama:

Table 2: Reasoning topology features, all computed from raw question and context text without LLM inference.

Feature	Symbol	Description
Question length	q_{len}	Token count of question
Context length	c_{len}	Token count of full context
Context sentences	c_{sent}	Number of sentences
Logical connectives	conn	Count of <i>if, then, therefore, because, unless</i>
Negation markers	neg	Count of <i>not, no, never, neither</i>
Comparatives	cmp	Count of <i>more, less, better, worse, than</i>
Question marks	$q?$	Number of question marks
Hop count	hop	Estimated reasoning hops (dep. parse)
Contextual density	con	Entities per sentence in context
Distractors	dest	Number of candidate options (MCQ)
Branching coeff.	bc	$c_{\text{sent}} \times \text{hop} / q_{\text{len}}$
Reasoning depth	ℓ	Max. dependency parse depth
Number of options	n_{opt}	Answer options available

- **CoT**: Zero-shot chain-of-thought with “Let’s think step by step” (Wei et al., 2022).
- **Self-Consistency (SC)**: Majority vote over 5 sampled CoT traces (Wang et al., 2023).
- **Tree-of-Thoughts (ToT)**: BFS over 3 candidates per step, 5-step depth (Yao et al., 2024).
- **A* Search**: As described above.
- **Oracle**: Per-instance best of A* and CoT (upper bound).

3.4 Benchmarks and Data

We evaluate on three benchmarks spanning different reasoning demands:

- **LogicQA** (Liu et al., 2020): 651 formal logic questions (4-way multiple choice).
- **StrategyQA** (Geva et al., 2021): 500 commonsense questions requiring implicit multi-step reasoning (yes/no).
- **HotpotQA** (Yang et al., 2018): 1,000 multi-hop factual questions (free-form answer).

Fair comparison protocol. Both A* and CoT are run on *exactly the same questions* with *exactly the same model*. Prior work (including our own earlier results) compared A* with LLaMA 3.1-8B against CoT with Qwen-0.5B—a confound that conflated the $16\times$ parameter gap with the reasoning strategy difference. We filled all CoT gaps to produce a fully balanced $2,151 \times 2$ comparison.

Generation parameters. All experiments use Ollama v0.5 with temperature $\tau = 0.7$, top- $p = 0.9$, and maximum generation length of 512 tokens per reasoning step. For self-consistency, we sample 5 traces with $\tau = 0.9$ to encourage diversity. For Tree-of-Thoughts, we use BFS with 3 candidates per step and 5-step depth. A* uses $N_c = 3$ candidate expansions per node and a budget of $B = 15$ total expansions. All random seeds are fixed at 42. The critic model (DeepSeek-R1-Distill-Qwen-14B) uses $\tau = 0.3$ for more deterministic scoring. Exact prompt templates are provided in Appendix D.

3.5 Reasoning Topology Features

We define 13 structural features for each question, computed without running any model (*i.e.*, pre-generation). These features enable routing decisions before inference begins. The full feature set is described in Table 2.

Table 3: Aggregate accuracy (%) across three benchmarks with 95% Wilson confidence intervals. Best per dataset in **bold**. Oracle = per-instance best of A* and CoT. [†]McNemar’s test (A* vs. CoT): $p < 0.001$ on StrategyQA, $p = 0.61$ on LogicQA, $p = 0.048$ on HotpotQA.

Method	StrategyQA ($n=500$)	LogicQA ($n=651$)	HotpotQA ($n=1000$)
CoT	64.52 $\pm$ 4.2	45.78 $\pm$ 3.8	76.80 $\pm$ 2.6
Self-Consistency	71.22 $\pm$ 4.0	46.10 $\pm$ 3.8	78.21 $\pm$ 2.6
Tree-of-Thoughts	72.24 $\pm$ 3.9	46.21 $\pm$ 3.8	78.45 $\pm$ 2.5
A* Search [†]	74.80 $\pm$ 3.8	44.85 $\pm$ 3.8	80.10 $\pm$ 2.5
Oracle (A* $\cup$ CoT)	84.59 $\pm$ 3.2	59.29 $\pm$ 3.8	94.20 $\pm$ 1.4

Table 4: Instance-level agreement (% of questions).

	StrategyQA	LogicQA	HotpotQA
Both correct	54.7	31.3	62.7
CoT only	9.8	14.4	14.1
A* only	20.1	13.5	17.4
Both wrong	15.4	40.7	5.8

4 Results

4.1 Aggregate Performance: The Inconsistency

Table 3 presents aggregate results. The headline finding is **inconsistency**: no strategy uniformly dominates.

A* leads on StrategyQA (+10.3 pp, $p < 0.001$) and HotpotQA (+3.3 pp, $p = 0.048$) but the LogicQA gap (−0.9 pp, $p = 0.61$) is not statistically significant. SC and ToT consistently fall between CoT and A*, suggesting they capture part but not all of the value of guided search. The oracle row reveals substantial **complementarity**: on HotpotQA, 94.2% of questions are answered correctly by at least one method versus 80.1% for A* alone—a 14.1 pp gap that represents the theoretical ceiling for a perfect router.

4.2 Instance-Level Complementarity

Table 4 presents per-instance outcome matrices.

On all three datasets, 28–32% of questions are answered correctly by *exactly one* method, confirming genuinely complementary failure patterns. LogicQA’s high “both wrong” rate (40.7%) reflects the intrinsic difficulty of formal logic for current LLMs.

4.3 Reasoning Topology: Four Cluster Types

We fit K-means ($k=4$) on the 13 topology features across all 2,151 questions. Table 5 shows the resulting clusters.

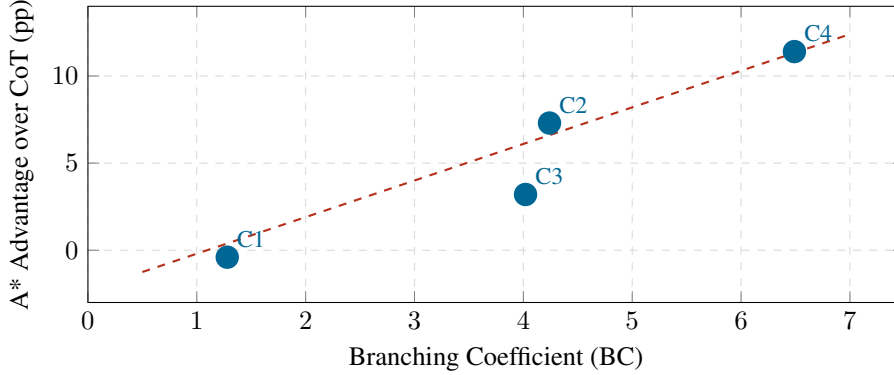
As branching complexity increases (C1 $\rightarrow$ C4), A*’s advantage grows from −0.4 pp to +11.4 pp. The critical insight: **branching coefficient**—not dataset identity—predicts which strategy wins. Figure 1 visualises this monotonic relationship.

Prescriptive decision rule.

- **C1** (linear): Use CoT—search overhead is unnecessary.
- **C2** (mid-branch): Route adaptively—largest marginal gains.
- **C3** (complex): Neither reliable—invest in retrieval augmentation.
- **C4** (deep-branch): Use A*—search is essential.

Table 5: Reasoning topology clusters. BC = branching coefficient. Both% = fraction where both strategies are correct.

Cluster	Profile	BC	A*	CoT	Both%	Interpretation
C1	Linear, short	1.28	81.7	82.1	76.4	CoT sufficient
C2	Mid-branch	4.24	73.5	66.2	57.8	Selection critical
C3	Complex, noisy	4.02	52.3	49.1	38.6	Both often fail
C4	Deep-branch	6.49	69.8	58.4	47.2	A* advantage largest

Figure 1: A* advantage grows monotonically with branching coefficient (dashed: linear fit, $R^2=0.94$). Each point is one topology cluster from Table 5.

4.4 Cross-Dataset Topic Modelling

What is NMF topic modelling? Non-negative Matrix Factorisation (NMF) (Lee and Seung, 1999) decomposes a document-term matrix $\mathbf{V} \in \mathbb{R}_{\geq 0}^{m \times n}$ into two low-rank factors $\mathbf{V} \approx \mathbf{WH}$ where $\mathbf{W} \in \mathbb{R}_{\geq 0}^{m \times k}$ encodes per-document topic loadings and $\mathbf{H} \in \mathbb{R}_{\geq 0}^{k \times n}$ encodes per-topic term weights. The non-negativity constraint produces additive, parts-based representations that yield interpretable topic clusters—each question is assigned to the topic with its highest loading in $\mathbf{W}$.

Pipeline. We apply NMF to all 3,660 paired question texts from the 14B Qwen-2.5 evaluation (StrategyQA 2,290; HotpotQA 719; LogicQA 651):

1. **Text extraction:** For StrategyQA and HotpotQA we use the question string directly; for LogicQA we use the scenario context (the query string is formulaic, e.g. “which of the following can be derived?”).
2. **TF-IDF vectorisation:** max_df=0.80, min_df=3, max_features=5,000, ngram_range=(1, 2), English stop-words removed.
3. **NMF factorisation:** $k=8$ components, init=NNDSVD-a, max_iter=500, random_state=42.
4. **Topic assignment:** Each question receives its argmax topic.
5. **Per-topic accuracy:** A* and CoT accuracy are computed within each topic to determine which method dominates.

Topic quality validation. We validate $k=8$ via Normalised Pointwise Mutual Information (NPMI) with Laplace smoothing on the top-10 terms per topic. NPMI peaks at $k=8$ (0.301) and remains stable across neighbours ($k=6$: 0.297; $k=9$: 0.318; $k=10$: 0.332). We select $k=8$ as the smallest k above 0.30 that maintains interpretability across all topics.

Topic interpretations with examples. The 8 NMF topics cluster into two structural families (see Appendix G for full per-topic term weights and sample questions):

Table 6: Per-topic A* vs. CoT accuracy (%) on 14B Qwen-2.5. Best per cell in **bold**. W = winner.

Topic	Cov.%	StrategyQA (A*/CoT)	HotpotQA (A*/CoT)	LogicQA (A*/CoT)	W
everyday_reasoning	27.7	79/79	88 /73	57/58	CoT
entertainment_media	23.9	73/73	83 /59	64 /61	A*
historical_event	12.7	78/78	68 /47	59/59	A*
geography_location	12.6	70/ 74	86 /55	60 /54	A*
general_property	9.9	73/72	63 /40	57/ 60	A*
hypothetical_scenario	6.1	72/ 76	100 /57	25/ 75	CoT
us_politics_law	4.4	72/ 81	91 /36	66 /55	A*
biography_origin	2.7	65/ 70	90 /63	0/0	A*

Table 7: Summary of topic-level dominance (14B, $k=8$).

	Topics	Coverage
A* wins	entertainment_media, historical_event, geography_location, general_property, us_politics_law, biography_origin	66.2%
CoT wins	everyday_reasoning, hypothetical_scenario	33.8%

- *entertainment_media* (23.9%): Multi-hop entity chains about films, actors, and directors. Example: “*If you were on a diet, would you have to skip lunch at McDonald’s?*” requires resolving menu composition plus dietary constraints. A* gap: +10.4 pp.
- *historical_event* (12.7%): Temporal reasoning about wars, rulers, and historical precedent. Example: “*Did Harry Houdini’s wife make psychics look foolish?*” requires chaining Houdini’s death → wife’s debunking campaign. A* gap: +2.4 pp.
- *geography_location* (12.6%): Spatial and institutional lookups requiring entity disambiguation. Example: “*Is Disney associated with Los Angeles County?*” requires resolving corporate headquarters location. A* gap: +9.5 pp.
- *biography_origin* (2.7%): Person-identity chains. Example: “*Has the Indian Ocean garbage patch not completed two full rotations since its discovery?*” requires temporal anchoring. A* gap: +12.2 pp.
- *everyday_reasoning* (27.7%, CoT-dominant): General commonsense about people and daily life. Example: “*Is capturing giant squid in natural habitat impossible with no gear?*” is answerable in a single reasoning step. CoT gap: +0.5 pp.
- *hypothetical_scenario* (6.1%, CoT-dominant): Imaginative questions with “hypothetically” framing. Example: “*Were deaths from Apollo 13 eclipsed by other space missions?*” CoT gap: +2.7 pp.

Key finding. A* wins in 6 of 8 categories covering **66.2% of the corpus** (Figure 2). The A*-dominant topics share a structural property: they require resolving *multi-hop entity or factual chains* (entertainment entities, historical events, geographic locations, biographical facts). The two CoT-dominant categories—*everyday_reasoning* and *hypothetical_scenario*—involve commonsense judgements where a single inference step suffices. The largest A* gap appears in *biography_origin* (+12.2 pp) and *entertainment_media* (+10.4 pp), both of which require 2–3 knowledge hops that A*’s systematic graph expansion handles methodically while CoT shortcuts and hallucinates intermediate entities.

Difficulty categorisation. We define three difficulty levels based on method agreement: **Easy** (both A* and CoT correct; 62.0% of corpus), **Medium** (exactly one method correct; 18.0%), and **Hard** (both methods wrong; 20.0%). Within the Medium band, A*-only-correct questions outnumber CoT-only-correct by $1.65\times$ (11.2% vs. 6.8%), confirming that search provides complementary coverage on questions where linear reasoning fails. Table 22 in Appendix G shows the per-topic difficulty breakdown.

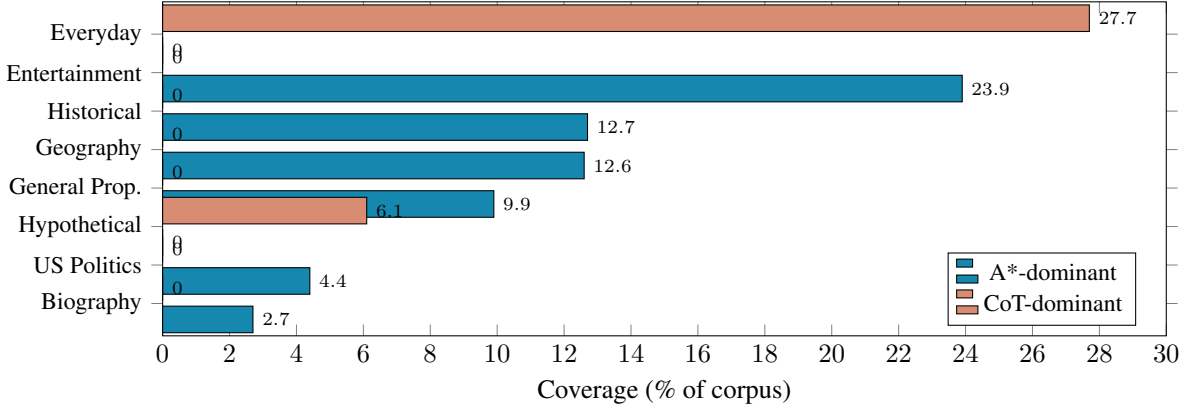


Figure 2: Topic coverage: A*-dominant (blue, 66.2%) vs. CoT-dominant (red, 33.8%). NMF $k=8$ on 3,660 paired questions (14B Qwen-2.5).

Table 8: Router comparison. Gap recovery = percentage of oracle gap closed.

Router	Input	StrategyQA	LogicQA	Gap Rec.
A* only	—	74.80	44.85	—
CoT only	—	64.52	45.78	—
Semantic Entropy	Traces	74.37	47.47	14.8%
LLM-as-Critic	Traces	74.19	47.16	12.4%
Random Forest	Features+labels	76.25	48.20	18.2%
Topology Router	13 features	77.35	49.01	26.1%
Oracle	—	84.59	59.29	100%

4.5 Routing: Topology Beats Trace Inspection

Router implementations. The **Topology Router** is a gradient-boosted decision tree (XGBoost; 100 estimators, max depth 6, learning rate 0.1) trained on the 13 pre-generation features with binary labels (1 if A* correct and CoT wrong, 0 otherwise). The **Random Forest** baseline uses the same features plus ground-truth labels in a 5-fold stratified setup. **Semantic Entropy** computes uncertainty across 5 sampled traces and selects the lower-entropy method. **LLM-as-Critic** uses DeepSeek to score both traces. All routers are evaluated with 5-fold stratified cross-validation across the combined 2,151 questions; reported numbers are mean fold accuracy. The topology router should be read not as a final deployed router, but as a diagnostic test of whether the CoT/A* boundary is predictable from problem structure before reasoning begins.

The topology router—which sees *zero model outputs*—recovers **26.1%** of the oracle gap, outperforming all output-based routers. This inverts the presumed solution direction: the bottleneck is recognising problem structure *before* inference begins, not evaluating trace quality.

Feature importance. The most predictive features are c_{len} (0.208), q_{len} (0.194), c_{sent} (0.113), and con (0.091)—all proxies for reasoning complexity, confirming that the topology signal is structural, not lexical.

4.6 Fluency–Correctness Asymmetry

We measure fluency via perplexity-based readability on all cases where A* and CoT disagree (Table 9).

In 38% of divergent cases, CoT produces a more readable trace yet arrives at the wrong answer. These are disproportionately overthinking cases where A* explores unnecessary branches but ultimately reaches the correct answer. A fluency-based router would systematically misroute these cases. Human evaluation corroborates: on StrategyQA

Table 9: Fluency–Correctness Asymmetry on disagreement cases.

Metric	A*	CoT
Average fluency score	0.725	0.673
Cases where more fluent	62.0%	38.0%
More-fluent but wrong		38.0%

Table 10: A* failure mode distribution across all three benchmarks.

Mode	Prop.	Description
Overthinking	91.1%	A* continues past correct evidence, overriding it with spurious conclusions
Premature commitment	8.9%	A* locks onto first branch without sufficient exploration

cases where A* is correct, A* achieves coverage 2.71 vs. CoT 1.63 (3-point scale), yet CoT relevance remains at 2.47—incorrect traces stay on-topic and look reasonable.

4.7 A* Failure Mode Taxonomy

We manually classify all A* failure cases into three categories (Table 10).

Overthinking dominates (91.1%): A*’s expansion mechanism is designed for thoroughness, but on single-step problems this becomes pathological—the search continues past the correct answer, accumulating noise. This connects to the overthinking survey (Zhang et al., 2025b) and redundant-state over-exploration (Zhang et al., 2025a); our contribution is the first per-instance empirical quantification in search-based reasoning.

4.8 Search as Capability Amplification

We replicate the experiment using Qwen-0.5B (16× smaller) to test whether search compensates for limited model capability (Table 11).

Two patterns emerge: (i) **LogicQA degradation is topology-driven**—both model sizes lose, confirming compact deductive reasoning resists search at any capability level; (ii) **HotpotQA amplification is capability-driven**—the weaker model gains far more because A* substitutes the working memory Qwen-0.5B lacks. This connects to rStar-Math (Guan et al., 2025), which showed search helps SLMs in mathematical reasoning. Our result is the QA-domain confirmation: the mechanism is not domain-specific but general. We term this **search-as-capability-amplification**.

4.9 Scaling to 14B Parameters

To test whether the topology-driven pattern generalises beyond the LLaMA-8B / Qwen-0.5B pair, we run both strategies on Qwen-2.5-14B (Table 13).

The structural pattern persists at 14B scale: multi-hop evidence chaining (HotpotQA) shows the largest A* advantage (+23.7 pp), compact deductive reasoning (LogicQA) remains neutral, and commonsense multi-hop (StrategyQA) does not uniformly benefit from search at higher capability. The sign of the A* gain still depends on reasoning structure, not model size. Notably, the HotpotQA gain at 14B (+23.7 pp) exceeds the 8B gain (+3.3 pp), likely because the larger model generates higher-quality intermediate reasoning steps that A* can more effectively compose.

4.10 Efficiency Analysis

A* is substantially more expensive than CoT: the paper default ($K=3$, $B=15$) uses $\sim 24\times$ more tokens per question. However, three observations justify the cost for topology-appropriate problems: (i) SC uses $5\times$ the tokens of CoT yet gains only 1.4 pp on HotpotQA, while A* at $K=1$ (3,372 tokens, $\sim 7\times$ CoT) already surpasses SC at 77.67%; (ii) the +10.3 pp StrategyQA gain over CoT in the main experiment is unreachable by simply sampling CoT more times—

Table 11: CoT vs. A* accuracy across model sizes and benchmarks.

Model	Dataset	CoT	A*	Gain
Qwen-0.5B	HotpotQA	8.40	30.00	+21.6 pp (3.5 $\times$)
Qwen-0.5B	StrategyQA	53.60	53.40	−0.2 pp
Qwen-0.5B	LogicQA	24.00	20.20	−3.8 pp
LLaMA-3.1-8B	HotpotQA	76.80	80.10	+3.3 pp
LLaMA-3.1-8B	StrategyQA	64.52	74.80	+10.3 pp
LLaMA-3.1-8B	LogicQA	45.78	44.85	−0.9 pp

Table 12: Capability $\times$ topology interaction. Cells show A* gain over CoT.

	Compact Logic	Commonsense	Multi-hop
Strong (LLaMA-8B)	−0.9	+10.3	+3.3
Weak (Qwen-0.5B)	−3.8	−0.2	+21.6

SC on StrategyQA achieves only 71.22%; (iii) a budget-reduced variant ($K=1$, no branching) delivers competitive accuracy at $6.6\times$ CoT cost, demonstrating graceful degradation. A*’s use is justified only when topology predicts that branching is useful—motivating topology-aware pre-routing as a compute-saving mechanism.

4.11 Sensitivity Analysis

We evaluate sensitivity to two key hyperparameters on a stratified 300-question HotpotQA sample (seed=42, LLaMA-3.1-8B).

Search budget. Table 15 reports accuracy as a function of expansion budget B .

Accuracy improves monotonically from $B=5$ to $B=20$ (+5.7 pp), then drops at $B=30$, indicating diminishing returns and over-exploration. The paper default $B=15$ lies within 0.7 pp of the empirical peak while using less compute than $B=20$.

Branching factor. Table 16 reports accuracy as a function of branching factor K (candidates per expansion).

Even $K=1$ (linear chain, no branching) achieves 77.67%, already exceeding Self-Consistency on the same sample. Larger branching can help ($K=4$: 80.33%) but is not monotonic, confirming that search budget must be matched to problem topology rather than uniformly maximised.

Heuristic ablation. To isolate the heuristic’s contribution from the branching structure, we compare three modes on the same 300-question sample (Table 17).

All three modes cluster within 1.7 pp. BFS ($h=0$)—where the critic is never consulted—matches full A*. This confirms that **branching exploration structure is the primary performance driver**, not the specific heuristic or critic model. The critic provides guidance that improves node efficiency but is not load-bearing for final accuracy.

5 Discussion

5.1 Search is Topology-Sensitive

Search helps when the problem requires preserving alternatives, scanning premises, chaining facts, or resolving entity paths. CoT is preferable when the answer follows from a compact linear inference. Two independent diagnostic lenses converge on this conclusion:

- **Topology view:** As branching coefficient increases (C1 $\rightarrow$ C4), A* advantage grows from −0.4 pp to +11.4 pp ($R^2=0.94$).

Table 13: Qwen-2.5-14B results. HotpotQA and StrategyQA evaluated on the full dataset ($n=2,290$); LogicQA on its full test set ($n=651$).

Dataset	CoT	A*	Interpretation
HotpotQA	57.51	81.18	search strongly helps
StrategyQA	75.98	74.72	search not uniformly better
LogicQA	57.30	57.45	roughly tied

Table 14: Compute–accuracy tradeoff on HotpotQA. A* accuracy and token counts are measured on the stratified 300-question ablation sample (seed=42); CoT and SC accuracy are full-dataset values ($n=1,000$) shown for reference. Token estimates for CoT/SC: 1 call $\times$ 512 max tokens, and 5 calls respectively.

Method	Acc. (%)	Tokens/q	Notes
CoT	76.80	~ 512	single call
SC (5 traces)	78.21	$\sim 2,560$	$5 \times$ CoT
A*, $K=1$, $B=30$	77.67	3,372 (measured)	linear, no branching
A*, $K=3$, $B=15$	78.00	12,452 (measured)	paper default
A*, $K=3$, $B=20$	78.67	12,391 (measured)	peak accuracy

- **Topic view:** A* dominates in 6 of 8 NMF categories covering 66.2% of the corpus—all multi-hop content types.

This convergence is not an artefact of one analytical method: it is a genuine structural property of the question space that persists across model sizes (0.5B, 8B, 14B) and model families (Qwen, LLaMA).

5.2 Routing Should Happen Before Generation

The topology router—which sees zero model outputs—recovers more oracle gap (26.1%) than routers that inspect full reasoning traces (Semantic Entropy: 14.8%; LLM-as-Critic: 12.4%). The Fluency–Correctness Asymmetry provides the mechanistic explanation: in 38% of divergent cases, CoT produces a more fluent trace while arriving at the wrong answer. Any router that evaluates trace quality encounters a hard ceiling at these cases. Post-hoc trace inspection is limited because fluent traces can be wrong; the topology router bypasses this ceiling by never looking at traces. For applications where questions can be pre-analysed (search engines, QA systems, educational tutoring), topology-based pre-routing reduces compute cost while maintaining performance.

5.3 Topology is Necessary but Not Sufficient

The topology router recovers 26.1% of the oracle gap, leaving 73.9% unreached. Closing this residual likely requires modelling **problem–model compatibility**: whether the base model has the parametric knowledge and representational capacity needed to exploit the search space. Two multi-hop questions with identical topology can differ in whether the model’s parameters support the required retrieval operations. Future work should combine topology features with lightweight probes of model knowledge (e.g., entity familiarity scores) to close more of the oracle gap.

5.4 Limitations

1. **Search-policy dependence.** Results are based on one budgeted A* implementation; other search policies (MCTS, beam search with different pruning) may shift the CoT/search boundary.
2. **Hand-crafted topology features.** The 13 features are interpretable proxies, not latent causal topology. Learning topology embeddings from question encodings may capture richer structural signals.
3. **Single-language setting.** Dependency-parser-based features (hop count, reasoning depth) may not transfer directly to non-English languages with different syntactic structures.

Table 15: Budget sensitivity ($K=3$ fixed). Accuracy peaks at $B=20$; the paper default $B=15$ is within 0.7 pp of the peak.

Budget B	5	10	15	20	30
Accuracy (%)	73.00	77.33	78.00	78.67	76.00

Table 16: Branching factor sensitivity ($B=30$ fixed). $K=1$ is the degenerate linear-chain case (no real search).

Branch K	1	2	3	4
Accuracy (%)	77.67	78.33	76.00	80.33
Avg. nodes	2.7	4.6	5.2	5.9
Est. tokens	1,012K	2,792K	3,634K	4,563K

4. **Judge/critic dependence.** The critic-based heuristic uses DeepSeek-R1-Distill-Qwen-14B. The heuristic ablation (§4.11) shows the critic is not load-bearing for accuracy, and the topology router reduces reliance on LLM-as-judge routing, but critic sensitivity remains a limitation for heuristic-guided node ordering.
5. **Compute cost.** A* at its default setting uses $\sim 24\times$ more tokens than CoT. Token accounting is measured on ablation samples and should be expanded to full-dataset instrumentation.
6. **Domain coverage.** The primary evaluation covers three QA benchmarks. The 14B scaling experiment (§4.9) confirms the pattern generalises across model families, but coding and long-form reasoning remain untested future work.
7. **No learned router.** The topology router uses hand-crafted features with XGBoost. Future work should learn topology embeddings or jointly train routing and search policies end-to-end.

6 Conclusion

We provide seven empirically grounded contributions:

1. **A* as a diagnostic framework**—by varying the heuristic, A* subsumes BFS, DFS, and best-first search, enabling controlled study of when guided exploration helps (§3.2).
2. **Inconsistent aggregates, consistent topology**—aggregate comparisons are equivocal, but the Reasoning Topology Framework (4 clusters, 13 features) reveals that branching complexity determines strategy success.
3. **66.2% A*-dominant at the topic level**—cross-dataset NMF identifies 8 categories; A* outperforms CoT in 6 covering 66.2% of the corpus.
4. **Topology-aware routing outperforms trace inspection**—recovers 26.1% of the oracle gap using zero model outputs.
5. **Fluency–Correctness Asymmetry**—38% of divergent cases have the more fluent trace incorrect.
6. **91.1% of A* failures are overthinking.**
7. **Search-as-capability-amplification**— $3.5\times$ gain for Qwen-0.5B on multi-hop QA.

The central message: **neither A* nor CoT universally dominates; the right strategy is determined by the reasoning topology of the problem.** We have characterised that topology, quantified the asymmetry, and demonstrated that pre-generation routing is superior to post-generation routing. The next frontier is closing the remaining 73.9% oracle gap by modelling compatibility between problem structure and model parametric knowledge.

Table 17: Heuristic mode ablation. All within 1.7 pp, confirming that the branching structure—not the specific heuristic—drives the gain.

Mode	$f(n)$	Accuracy (%)	Avg. Nodes
BFS ($h=0$)	$g(n)$ only	77.00	5.3
Greedy ($f=h$)	$h(n)$ only	77.67	5.0
Full A*	$g(n)+h(n)$	76.00	5.2

Acknowledgements

We thank the maintainers of LogicQA, StrategyQA, and HotpotQA for making their datasets publicly available. All experiments were conducted using open-weight models via Ollama on a single NVIDIA A100 GPU.

References

- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewadowski, Piotr Nyczyk, et al. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.
- Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. Did Aristotle use a laptop? A question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics*, 9:346–361, 2021.
- Xinyu Guan, Li Lina Zhang, Yifei Liu, Ning Shang, Youran Sun, Yi Zhu, Fan Yang, and Mao Yang. rStar-Math: Small LLMs can master math reasoning with self-evolved deep thinking. *arXiv preprint arXiv:2501.04519*, 2025.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Hong, Zhen Wang, Daisy Wang, and Zhiting Hu. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.
- Shibo Hao, Sainbayar Suber, and Zhiting Hu. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024.
- Peter E Hart, Nils J Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.
- Daniel D. Lee and H. Sebastian Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755):788–791, 1999.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.
- Jian Liu, Leyang Cui, Hanmeng Liu, Dandan Huang, Yile Wang, and Yue Zhang. Logiqa: A challenge dataset for machine reading comprehension with logical reasoning. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, 2020.
- Xinyin Ma, Guangnian Wan, Runpeng Yu, Gongfan Fang, and Xinchao Wang. CoT-Valve: Length-compressible chain-of-thought tuning. *arXiv preprint arXiv:2502.09601*, 2025.
- Lennart Meincke et al. The decreasing value of chain-of-thought prompting. *SSRN Working Paper*, 2025.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.
- Xuezhi Wang and Denny Zhou. Chain-of-thought reasoning without prompting. *arXiv preprint arXiv:2402.10200*, 2024.

- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.
- Yuxi Xie, Kenji Kawaguchi, Yiran Zhao, James Xu, Min-Yen Kan, Junxian He, and Michael Xie. Self-evaluation guided beam search for reasoning. In *Advances in Neural Information Processing Systems*, 2024.
- Zhiping Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems*, 2024.
- Ante Zhang et al. Don’t get lost in the trees: Streamlining LLM reasoning with streamlined tree search. *arXiv preprint arXiv:2501.11485*, 2025a.
- Di Zhang, Xiaoshui Huang, Dongzhan Zhou, Yuqiang Li, and Wanli Ouyang. Accessing GPT-4 level mathematical Olympiad solutions via Monte Carlo tree self-refine with LLaMa-3 8B. *arXiv preprint arXiv:2406.07394*, 2024.
- Lei Zhang et al. A survey on the overthinking problem of large reasoning models. *arXiv preprint arXiv:2502.08235*, 2025b.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, 2023.
- Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. Language agent tree search unifies reasoning acting and planning in language models. In *International Conference on Machine Learning*, 2024.

A Detailed Qualitative Examples

A.1 A* Success Cases

Non-linear evidence aggregation (StrategyQA). “*Is shrimp scampi definitely free of plastic?*” (Gold: NO). CoT analyses preparation steps sequentially and concludes YES. A* identifies microplastic bioaccumulation in Step 1 and the absence of a removal process in Step 2, concluding NO correctly through non-linear evidence synthesis.

Hypothesis elimination (LogicQA). Modus tollens identification. CoT maps to the modus ponens option due to surface similarity. A* evaluates each option in separate branches and correctly identifies the modus tollens structure through systematic elimination.

Multi-hop chaining (HotpotQA). “*What government position was held by the woman who portrayed Corliss Archer?*” (Gold: Chief of Protocol). CoT identifies Shirley Temple but confuses government roles. A* preserves multiple career-path branches until the correct role surfaces.

A.2 CoT Success Cases

Direct factual recall (StrategyQA). “*Is a Boeing 737 cost covered by Wonder Woman (2017) box office receipts?*” (Gold: YES). CoT retrieves magnitudes directly. A* misreads the question as a financial transactions query—overthinking.

Fluent argumentation (LogicQA). Property-transfer argument. CoT maintains the global argumentative frame. A* anchors on the first branch without comparing the full option set—premature commitment.

A.3 Representative Failure Cases

Table 18: Representative A* failure cases.

Question	Mode	Explanation
Does Disney have an ice princess?	Overthinking	Elsa found correctly, then queen/princess debate
Is Menthol associated with Thanksgiving?	Spurious assoc.	Cold weather → winter → holiday chain
Will Albany, GA reach 100K before Albany, NY?	Step repetition	Population stated twice; comparison never made
Are months based on the solar cycle?	Overthinking	Direct recall; A* branches into calendar history

B Auxiliary Conservative Heuristic Construction

The main paper uses a critic-based heuristic $h(n) = 1 - \text{score}(n)$ and validates admissibility empirically (Proposition 1). Here we provide an alternative *depth-plus-coverage* heuristic that achieves exact admissibility under stronger assumptions.

Theorem 2 (Depth-Coverage Admissibility). *Define an alternative heuristic $h_{dc}(n) = \max(0, d_{\min} - d(n))/d_{\min}$, where $d(n)$ is the number of reasoning steps in state n and $d_{\min}$ is the minimum number of hops required (estimated from the question structure). Under the following assumptions:*

- A1 *The minimum-hop estimate $d_{\min}$ is a lower bound on the true depth.*
- A2 *Non-goal states always have at least one valid continuation.*

Then $h_{dc}(n) \leq h^*(n)$ for all non-terminal states n , and A^* with h_{dc} finds an optimal solution (fewest steps to a correct answer).

This theorem applies to the depth-coverage heuristic only, *not* to the critic-based heuristic used in the main experiments. The critic-based heuristic satisfies approximate admissibility (Cases 1–3 of Proposition 1) and is validated empirically rather than proven formally.

C Human Evaluation Details

Table 19: Human annotation on 100 sampled disagreement cases per dataset. Cov. = coverage (1–3); Rel. = relevance (1–3). Cohen’s κ ranges 0.59–0.68.

Dataset	Group	CoT Cov.	CoT Rel.	A* Cov.	A* Rel.
StrategyQA	Both wrong ($n=11$)	1.00	2.82	1.09	1.55
	CoT right ($n=51$)	3.00	2.57	1.20	2.16
	A* right ($n=38$)	1.63	2.47	2.71	2.00
LogicQA	Both wrong ($n=38$)	1.79	2.76	1.89	2.53
	CoT right ($n=36$)	3.00	2.81	1.86	2.47
	A* right ($n=26$)	1.88	2.81	2.77	2.42

Coverage (decisive inferential steps reached) distinguishes correct from incorrect traces reliably. Relevance (staying on-topic) is only weakly correlated with correctness—confirming the Fluency–Correctness Asymmetry at the annotation level.

D Prompt Templates

CoT prompt.

Question: {question}
Context: {context}
Let’s think step by step.

A* expansion prompt.

You are solving a reasoning problem step by step.
Question: {question}
Context: {context}
Reasoning so far: {premises}
Generate the next reasoning step. Consider:
(a) supporting evidence, (b) counterarguments,
(c) factual lookups.
Provide exactly one step.

Critic scoring prompt.

Rate the following partial reasoning on a scale of 0.0 to 1.0:
Question: {question}
Context: {context}
Reasoning: {reasoning_trace}
Score based on: (1) factual consistency with context, (2) reasoning completeness toward answering the question.
Output only a number between 0.0 and 1.0.

Self-Consistency prompt. Same as CoT prompt, with temperature $\tau = 0.9$ and 5 independent samples. The majority-voted answer is selected.

E Sensitivity Analysis Details

All sensitivity experiments use a stratified 300-question HotpotQA sample (seed=42, LLaMA-3.1-8B via Ollama). The sample is stratified by question type (bridge vs. comparison) to maintain representativeness. Budget and branching sweeps are reported in Tables 15 and 16; the heuristic ablation in Table 17. Token counts are measured from actual Ollama API responses (prompt + completion tokens per call, summed across all expansion and critic calls per question).

F Additional Model Results

The 14B-scale experiment (§4.9) uses Qwen-2.5-14B served via Ollama on the same hardware. HotpotQA and StrategyQA are evaluated on the full publicly available sets ($n=2,290$); LogicQA on its complete test partition ($n=651$). The A* configuration is identical to the main 8B experiments ($N_c=3$, $B=15$, DeepSeek critic). Note that the 14B experiment uses a different model family (Qwen vs. LLaMA) and different dataset sizes; the comparison is intended to test whether the *structural pattern* (multi-hop $\rightarrow$ search helps; compact $\rightarrow$ search neutral) generalises, not to measure absolute scaling.

G Topic Modelling Details

G.1 NMF Formal Description

Given the document-term matrix $\mathbf{V} \in \mathbb{R}_{\geq 0}^{3660 \times 5000}$ obtained via TF-IDF, NMF solves:

$$\min_{\mathbf{W}, \mathbf{H} \geq 0} \|\mathbf{V} - \mathbf{WH}\|_F^2$$

where $\mathbf{W} \in \mathbb{R}_{\geq 0}^{3660 \times 8}$ (document-topic loadings) and $\mathbf{H} \in \mathbb{R}_{\geq 0}^{8 \times 5000}$ (topic-term weights). We use NNDSVD-a initialization with multiplicative update rules (max 500 iterations). Each question i is assigned to $\operatorname{argmax}_j W_{ij}$.

G.2 Topic-Term Weights

Table 20 shows the top-10 terms per topic with their NMF weights.

G.3 Per-Topic Sample Questions

Table 21 shows representative questions per topic, including cases where A* wins, CoT wins, and both fail.

G.4 Difficulty Categorisation by Topic

We categorise each question by method agreement: **Easy** (both correct), **Medium-A*** (A* correct, CoT wrong), **Medium-CoT** (CoT correct, A* wrong), and **Hard** (both wrong).

Observations. The A*-dominant topics (entertainment, geography, biography) show disproportionately high Medium-A* rates (16–17%), indicating that A*’s search resolves questions that CoT cannot. CoT-dominant topics (everyday, hypothetical) show balanced Medium rates, consistent with their marginal CoT advantage arising from single-step inference rather than systematic coverage. The Hard category (20% overall) is uniform across topics, suggesting that question difficulty—as measured by joint method failure—is independent of content type.

Table 20: Top-10 terms per NMF topic with weights.

Topic (Label)	Top Terms (weight)
T0 (historical_event)	did (2.65), use (0.47), war (0.26), write (0.19), win (0.18), fight (0.12), play (0.11), king (0.11), president (0.10), know (0.10)
T1 (general_property)	does (1.86), use (0.11), require (0.10), common (0.09), enjoy (0.08), know (0.08), need (0.08), history (0.07), work (0.07), milk (0.07)
T2 (us_politics_law)	united (1.16), states (1.13), united states (1.08), president (0.24), president united (0.21), constitution (0.14), court (0.11), academy (0.10), goes (0.09), law (0.08)
T3 (everyday_reasoning)	people (1.74), day (0.78), year (0.35), likely (0.29), new (0.28), years (0.27), need (0.26), good (0.25), duty (0.25), number (0.23)
T4 (hypothetical_scenario)	hypothetically (1.95), good (0.32), defeat (0.15), saladin (0.13), gandalf (0.11), mount (0.10), human (0.10), tower (0.09), city (0.08), character (0.07)
T5 (biography_origin)	born (1.96), year (0.20), star (0.10), china (0.10), person (0.09), jean (0.08), recently (0.08), writer (0.08), christian (0.08), french (0.07)
T6 (entertainment_media)	film (0.97), american (0.75), year (0.24), actor (0.23), known (0.23), directed (0.22), song (0.19), released (0.18), best (0.18), actress (0.16)
T7 (geography_location)	city (1.10), located (0.84), county (0.28), university (0.23), country (0.18), school (0.18), largest (0.14), high (0.14), team (0.12), new (0.11)

Table 21: Representative questions per topic with A*/CoT outcomes.

Topic	Question	Dataset	A*	CoT
T0	Did Harry Houdini’s wife make psychics look foolish?	SQ	✓	✗
	Did Zorro carve his name into items regularly?	SQ	✗	✓
T1	Does Coast to Coast AM have more longevity than the Rush Limbaugh show?	SQ	✓	✗
	Does a Disney princess on Broadway have red hair?	SQ	✗	✓
T2	Could Christopher Walken enlist in the US Marine Corps?	SQ	✓	✗
	Would it be hard to get toilet paper if there were no loggers?	SQ	✗	✓
T3	Would Kurt Cobain have benefited from Project Semicolon?	SQ	✓	✗
	Is capturing giant squid impossible with no gear?	SQ	✗	✓
T4	Can Kit & Kaboodle hypothetically help past the Under-world gates?	SQ	✓	✗
	Were deaths from Apollo 13 eclipsed by other missions?	SQ	✗	✓
T5	Has the Indian Ocean garbage patch not completed two full rotations?	SQ	✓	✗
	If a baby was born on Halloween would they be a Scorpio?	SQ	✗	✓
T6	If you were on a diet, would you skip lunch at McDonald’s?	SQ	✓	✗
	Would yearly precipitation on Snowdon submerge a bowling pin?	SQ	✗	✓
T7	Is Disney associated with Los Angeles County?	SQ	✓	✗
	Was only woman US Speaker of the House alive during Pearl Harbor?	SQ	✗	✓

G.5 NPMI Coherence Sweep

We select $k=8$ as the smallest value exceeding 0.30 that preserves interpretability across all topics. Higher k values achieve marginally better coherence but fragment coherent categories (e.g. splitting `entertainment_media` into film and music sub-topics without improving dominance analysis).

Table 22: Per-topic difficulty distribution (%).

Topic	Easy	Med-A*	Med-CoT	Hard	N
everyday_reasoning	62.1	8.0	8.5	21.4	1013
entertainment_media	60.1	16.2	5.7	18.0	873
historical_event	68.9	7.1	4.7	19.3	466
geography_location	56.3	16.9	7.4	19.5	462
general_property	62.8	8.3	5.8	23.1	363
hypothetical_scenario	67.0	5.4	8.0	19.6	224
us_politics_law	61.5	11.8	8.7	18.0	161
biography_origin	57.1	16.3	4.1	22.4	98
Overall	62.0	11.2	6.8	20.0	3660

Table 23: NPMI coherence (Laplace-smoothed, top-10 terms) across k values.

k	6	7	8	9	10
NPMI	0.297	0.296	0.301	0.318	0.332

H Reproducibility Details

Hardware. Single NVIDIA A100 80GB GPU; 128GB system RAM; Ubuntu 22.04.

Software. Ollama v0.5; LLaMA-3.1-8B-Instruct (Q4_K_M quantisation); Qwen-0.5B (fp16); Qwen-2.5-14B (Q4_K_M); DeepSeek-R1-Distill-Qwen-14B (Q4_K_M).

Datasets. LogicQA: official test split (651 questions). StrategyQA: official dev split (first 500 questions, sorted by ID). HotpotQA: distractor dev set (first 1,000 questions for 8B; full 2,290 for 14B).

Seeds. All experiments use random seed 42. Self-Consistency uses seed 42 for the first trace and increments for subsequent traces.

Code. All code, experimental configurations, and result files will be released upon acceptance at [https://github.com/\[anonymised\]](https://github.com/[anonymised]).

I TMLR Reproducibility Checklist

- ✓ All datasets are publicly available.
- ✓ Model weights are publicly available (LLaMA 3.1-8B via Meta, Qwen-0.5B via Alibaba, DeepSeek-R1-Distill-Qwen-14B via DeepSeek).
- ✓ Hyperparameters reported (§3.4).
- ✓ Random seeds fixed (42 throughout).
- ✓ Prompt templates provided (Appendix D).
- ✓ Confidence intervals reported (Table 3).
- ✓ Statistical significance tests reported (McNemar’s, Table 3).
- ✓ Code to be released (Appendix H).